BLUEPRINTS - UNREAL

SCENARIO 3: CLASS BLUE PRINT – ENABLE DISABLE LIGHT

ENABLE/DISABLE LIGHT USING AN INPUT KEY – USING A CLASS BLUE PRINT

ITEMS NEEDED FOR THIS OBJECTIVE

•

New level with floor - third person

•

Starter content

walls with door

walls

ceiling

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CREATING A BLUE PRINT CLASS

Content browser - Go to Blue print Folder – right click to Create Blue Print Class

or go to Blue print in the menu and create class blueprint –

Select the any one parent classes

Select Actor – then the panel will show up – to see where it is going to be saved – locate it in the content browser – blue prints folder –

Let‟s give a name for the blue print –"Wall blueprint " - Save All

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SETTING UP CLASS BLUE PRINT

Go to the Content browser and double click the blue print that you have saved. A new class blue print panel opens up – it has 3 modes – default, components and graph mode

Open Blue Print Class Panel and Content Browser Side by Side – From the CB – Look for Props

-

SM_Lamp_Wall – Drag from the CB and Drop it inside the Component view of the Blue Print Class – White Sphere is the Root – Drag the SM_Lamp_Wall on top of the Root Node

NAVIGATING THE VIEWPORT – ALT + LMB to Orbit, ALT + RMB - Dolly, W, A, S, D or Mouse Wheel- and Click Compile –

Drag a Copy from the Content Browser – Blue Print Class – Light to the Perspective Viewport – Align it Against the Wall

" What you see in the Viewport is Class Blue Print (Prefab – unity) "

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ADDING LIGHTS IN COMPONENT MODE (BP CLASS PANEL) – Click Add Component – Add a Spot Light – Move/Rotate to Put the Spot light in the Mesh – As you adjust – You will see the Light moving in the Perspective Viewport as well – Adjust the Cone Angle and the Lights to get a Proper Light Setup .

Note – Add a Point Light (Bounce Light) with low intensity to get more ambient lighting for the Lamp.

ADDING INTERACTION

ADDING TRIGGER VOLUME

In the BP Class Panel – Add Component – Select Box Collision – Drag it to the Viewport of BP – Scale it so that the Collider can be slightly big – Let‟s rename it Trigger Volume In the Perspective Viewport – You will see the Trigger Volume surrounding it

ADDING TEXT VOLUME

Add a Text Component – Text Render – Go to Details Panel do the changes.

•

Horizontal Alignment – Center

•

Pull the Text away from Light – Rotate if needed and place it in a Position where a player can read it

•

Change the Text to " Press F to Toggle " (change in the Details Panel – Text)

SAVE NOW THE BLUE PRINT AND THE MAP AS WELL

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ADDING FUNCTIONALITY

Go to BP – Event Graph

Select the Trigger Volume – from the Component Window – Go to Event Graph – Add the Two Events as Shown – Begin Overlap and End Overlap

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Select the Spot Light – Go to Details Panel – uncheck Rendering Visible

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Drag the Spot Light from the Component to the Event Graph – Options – Get or Set CTRL + Drag Light will drop in a Get button of the Component , ALT + Drag Light will drop in a Set Version of the Component – For this Example – will be using only the Get Version of the Component

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CTRL+Drag the Light into the Event Graph – Drag a Connector(a Small Blue Circle) Type in Toggle Visibility

•

Connect the graph as show

•

Copy the Lights in Different Locations – And you will see the Control Mechanism works for all Lights (On or Off the Light)

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ENABLING INPUT FUNCTIONALITY

Before we add input – Go to the Event Graph – Disconnect the Nodes as shown –

The Only way we can receive input is through Player Controller-

1)

Go to Begin Overlap Node – Drag the wire out and type in Enable Input -

2)

Drag a wire from player controller and type in get player controller

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DISABLING INPUT

1)

Go to End Overlap Node – Drag the wire out and type in Disable Input -

2)

Drag a wire from player controller and type in get player controller

In the Empty Space in the Event Graph – Right Click and Press „F" – You should have the Input Mappings of the F Key – Drag it to the Event Graph and Connect the Nodes as shown.

Now when the Player Enters the Trigger Zone – the Key Input is Enabled – When it Leaves the Trigger zone – the Key Input is Disabled

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DISPLAY MESSAGES WHEN INPUT IS ENABLED

CTRL + drag the Helpful message from the Add Component window to the Event Graph Drag a Wire and connect it to Toggle Visibility

Drag the Wire from the Enable/Disable Input

D:\UEProjects\BlueprintLesson\BlueprintLesson\Content\ToggleLight.umap

assist me to do this tutorial

+ Thought: Providing tutorial assistance · 5.8s

+ Thought: Clarifying input setup · 967ms

1. Open/Create the level

2. Open D:\UEProjects\BlueprintLesson\BlueprintLesson\Content\ToggleLight.umap (or create a new Third Person level with a floor), then Save All.

3. Make sure Starter Content is enabled (you can use StarterContent/Props/SM_Lamp_Wall).

4. Create the Class Blueprint

5. Content Browser → your Blueprints folder → Right click → Blueprint Class.

6. Pick Actor.

7. Name it BP_WallLampToggle (any name is fine) → Save.

8. Add components (Components tab)

9. Add → Static Mesh (or drag SM_Lamp_Wall into Components) and attach under DefaultSceneRoot.

- Set Static Mesh = SM_Lamp_Wall.

10. Add → Spot Light (name it SpotLight_Main) and position it inside the lamp.

- Optional: adjust Inner/Outer Cone Angle.

- For the tutorial: in Details, uncheck Visible (start OFF).

11. Optional bounce: Add → Point Light (name PointLight_Bounce), low intensity, also set Visible = false initially.

12. Add → Box Collision (name TriggerVolume), scale it so the player can stand “in front of” the lamp.

13. Add → Text Render (name HelpText)

- Text: Press F to Toggle

- Horizontal Alignment: Center

- Move it so the player can read it.

- Set Visible = false initially (so it only shows in the trigger).

14. Wire the interaction (Event Graph)

15. Select TriggerVolume in Components → in Event Graph, right click → add:

- OnComponentBeginOverlap (TriggerVolume)

- OnComponentEndOverlap (TriggerVolume)

16. Filter to player (recommended):

- From Other Actor → Cast To BP_ThirdPersonCharacter (or your character class).

- Only continue if the cast succeeds (do this for both Begin/End overlap).

17. Begin Overlap chain

- Enable Input

- Player Controller pin: Get Player Controller (0)

- Also show the message: HelpText -> Set Visibility (true)

18. End Overlap chain

- Disable Input

- Player Controller pin: Get Player Controller (0)

- Hide the message: HelpText -> Set Visibility (false)

19. Input key

- Right click empty graph → add Keyboard F (Input Key F).

- From Pressed:

- SpotLight_Main -> Toggle Visibility

- (Optional) PointLight_Bounce -> Toggle Visibility

20. Compile / Place / Test

21. Compile the Blueprint.

22. Drag multiple copies of BP_WallLampToggle into the level, align to walls.

23. Press Play:

- Walk into the box: text appears, F works.

- Walk out: text hides, F stops working.

If you tell me your Unreal version (UE4 vs UE5) and whether your project uses Enhanced Input, I can match the exact nodes/menu names you'll see for the F input.

▣ Build · GPT-5.2 · 18.1s

why when i place the lamp on the wall it is going into the wall. inspect and let me know

+ Thought: Inspecting in Unreal · 3.2s

→Skill "unreal-mcp"

I'm going to query your Unreal Editor via MCP for the selected lamp actor and its component transforms/pivot clues. Please select the placed BP_WallLampToggle instance (or the SM_Lamp_Wall actor) in the viewport/World Outliner, then I'll inspect it.

⚙unreal_unreal_ping

⚙unreal_unreal_get_editor_status

⚙unreal_unreal_get_engine_version

⚙unreal_unreal_get_selected_actors

⚙unreal_unreal_get_component_tree

+ Thought: Analyzing component tree · 6.7s

+ Thought: Adjusting lamp placement · 2.1s

+ Thought: Recommending adjustments · 1.9s

⚙unreal_unreal_get_open_editors

⚙unreal_unreal_get_active_blueprint

⚙unreal_unreal_get_blueprint_component_tree [object_path=/Game/BP_ToggleLight.BP_ToggleLight]

+ Thought: Investigating blueprint tools · 2.0s

⚙unreal_unreal_inspect_blueprint [object_path=/Game/BP_ToggleLight.BP_ToggleLight, include_cdo_properties=false]

+ Thought: Optimizing asset placement · 10.1s

+ Thought: Adjusting mesh offsets · 3.2s

+ Thought: Setting up component hierarchy · 1.9s

+ Thought: Inspecting object properties · 886ms

⚙unreal_unreal_inspect_object [actor_name=BP_ToggleLight_C_1, include_transient=false, max_properties=2000, name_contains=Collision]

From inspecting your placed actor BP_ToggleLight_C_1, the root component is SM_Lamp_Wall (your Static Mesh). That means the actor's placement pivot in the level is the mesh's pivot. If SM_Lamp_Wall's pivot is not on the backplate (it's often closer to the middle), placing it "on" a wall will put part of the mesh inside the wall.

Fix (recommended, doesn't require editing the mesh asset):

1. Open BP_ToggleLight.

2. Components tab: Add Component → Scene (name it Root).

3. Make Root the root component (drag it to the top, or right-click → make root).

4. Attach SM_Lamp_Wall under Root.

5. Select SM_Lamp_Wall and set Relative Location along the axis that points out from the wall (usually X) until it sits flush (start with X = 5 then adjust).

6. Keep SpotLight, Box, HelpText attached under SM_Lamp_Wall so they move with it.

Also worth checking (secondary cause): if SM_Lamp_Wall collision is NoCollision, the editor won't "stop" it from intersecting walls while you drag/place it.

▣ Build · GPT-5.2 · 1m 5s

lamp mesh origin -> should the x y or z be facing out of the wall

In Unreal the "out of the wall" direction is the mesh's forward axis.

Default convention is:

- X+ (red) = forward (should face out from the wall)

- Y+ (green) = right

- Z+ (blue) = up

So for a wall lamp, you typically want the lamp's X+ arrow pointing away from the wall, and you push/pull it flush by adjusting Relative Location X (after you add a Scene root as described).

If your lamp's forward is not X+ (some meshes are authored with Y+ forward), then use whatever axis the lamp points along as "out", but keep the Blueprint consistent: rotate the mesh component so its forward (usually X+) faces out, then position it.

▣ Build · GPT-5.2 · 7.0s

now green is left

That's normal. The gizmo colors are fixed axes, but whether green (Y) looks like "left" depends on:

- Local vs World space (gizmo setting)

- The actor's rotation

- Which direction you're currently looking from

In Unreal's default coordinate system:

- X+ (red) = Forward

- Y+ (green) = Right when you are facing X+

- Z+ (blue) = Up

If you rotate the lamp 180° (so it faces the opposite way), Y+ will appear on the other side relative to the camera, so “green is left” doesn’t mean it’s wrong.

To make this unambiguous for wall placement:

1. In the viewport, set the transform gizmo to Local (not World).

2. Rotate the lamp so the red arrow (X+) points out from the wall.

3. Then move it slightly along local X to sit flush (small X offset).

▣ Build · GPT-5.2 · 7.6s

Objective

- Build an Unreal Engine Blueprint Class that toggles a wall lamp light on/off with the F key, enabled only while the player is inside a trigger volume.

- Fix placement issue where the wall lamp blueprint instance sinks into/intersects the wall when placed.

Important Details

- UE version detected: 5.6.1-44394996+++UE5+Release-5.6

- Level/map referenced: D:\UEProjects\BlueprintLesson\BlueprintLesson\Content\ToggleLight.umap (also seen in editor as /Game/ToggleLight)

- Blueprint used in editor: /Game/BP_ToggleLight (generated class BP_ToggleLight_C)

- Placed instance inspected: BP_ToggleLight_C_1

- Placement issue likely due to mesh pivot/root setup: root component is the mesh SM_Lamp_Wall, so actor pivot = mesh pivot, causing wall intersection

- Secondary note: if SM_Lamp_Wall collision is NoCollision, placement won’t be constrained by wall collision while dragging

Work State

Completed

- Tutorial steps outlined to create BP_WallLampToggle-style actor: add SM_Lamp_Wall, Spot Light, optional Point Light, Box Collision trigger, Text Render prompt, overlap events to enable/disable input, F key to toggle light visibility and show/hide text

- MCP inspection performed:

- Selected actor: BP_ToggleLight_C_1

- Component tree fetched: SM_Lamp_Wall (root), SpotLight parented to mesh, Box parented to mesh, HelpText parented to mesh

- Open editor asset: BP_ToggleLight at /Game/BP_ToggleLight

- Attempted unreal_unreal_get_blueprint_component_tree returned error: Unknown method

Active

- Diagnosing lamp going into wall; recommended fix drafted: add a Scene root and offset SM_Lamp_Wall via relative location so it sits flush on the wall

Blocked

- (none)

Next Move

1. In BP_ToggleLight, add Scene component Root, make it the root, attach SM_Lamp_Wall under it, and adjust SM_Lamp_Wall relative offset (typically +X) until flush with wall.

2. Verify SM_Lamp_Wall collision settings (and wall collision) if placement snapping still allows intersection.

Relevant Files

- D:\UEProjects\BlueprintLesson\BlueprintLesson\Content\ToggleLight.umap: target level where the blueprint is placed/tested

- /Game/BP_ToggleLight: blueprint class implementing trigger + F-key toggle logic

- /Game/ToggleLight (map asset path seen in editor): contains placed instance BP_ToggleLight_C_1

▣ Compaction · GPT-5.2 · 8.5s

check again -> why when i am near the lamp, an dpress F it doesnt light up?

+ Thought: Diagnosing Unreal Engine Issue · 4.3s

Checking the BP_ToggleLight blueprint wiring (overlap + Enable Input + F key) and whether the lamp’s light component is being toggled/has intensity. I’ll pull the current blueprint + compile messages from the editor via MCP.

→Skill "unreal-mcp"

⚙unreal_unreal_ping

⚙unreal_unreal_get_editor_status

⚙unreal_unreal_get_current_project

⚙unreal_unreal_get_selected_actors

⚙unreal_unreal_get_open_editors

⚙unreal_unreal_get_active_blueprint

⚙unreal_unreal_get_component_tree

⚙unreal_unreal_inspect_blueprint [object_path=/Game/BP_ToggleLight.BP_ToggleLight, include_cdo_properties=true, max_properties=400]

⚙unreal_unreal_get_blueprint_graphs [object_path=/Game/BP_ToggleLight.BP_ToggleLight]

⚙unreal_unreal_get_compile_messages [object_path=/Game/BP_ToggleLight.BP_ToggleLight]

⚙unreal_unreal_get_blueprint_variables [object_path=/Game/BP_ToggleLight.BP_ToggleLight]

⚙unreal_unreal_get_blueprint_functions [object_path=/Game/BP_ToggleLight.BP_ToggleLight]

⚙unreal_unreal_get_blueprint_metadata [object_path=/Game/BP_ToggleLight.BP_ToggleLight]

+ Thought: Investigating blueprint methods · 4.2s

⚙unreal_unreal_get_blueprint_graph [object_path=/Game/BP_ToggleLight.BP_ToggleLight, graph=EventGraph, mode=compact, include_pins=false, include_edges=false, max_nodes=400]

⚙unreal_unreal_find_blueprint_events [object_path=/Game/BP_ToggleLight.BP_ToggleLight, all_graphs=true, limit=100]

⚙unreal_unreal_find_blueprint_nodes [object_path=/Game/BP_ToggleLight.BP_ToggleLight, query=Enable Input, all_graphs=true, limit=50]

⚙unreal_unreal_find_blueprint_nodes [object_path=/Game/BP_ToggleLight.BP_ToggleLight, query=Disable Input, all_graphs=true, limit=50]

⚙unreal_unreal_find_blueprint_nodes [object_path=/Game/BP_ToggleLight.BP_ToggleLight, query=F, all_graphs=true, limit=80]

⚙unreal_unreal_find_blueprint_nodes [object_path=/Game/BP_ToggleLight.BP_ToggleLight, query=Toggle, all_graphs=true, limit=80]

+ Thought: Adjusting blueprint graph query · 2.1s

⚙unreal_unreal_get_blueprint_graph [object_path=/Game/BP_ToggleLight.BP_ToggleLight, graph=EventGraph, mode=compact, include_edges=true, include_pins=false, max_nodes=400, max_edges=800]

⚙unreal_unreal_inspect_object [actor_name=BP_ToggleLight_C_1, include_transient=false, max_properties=600]

+ Thought: Troubleshooting collision settings · 2.8s

⚙unreal_unreal_inspect_object [object_path=/Game/ToggleLight.ToggleLight:PersistentLevel.BP_ToggleLight_C_1.Box, include_transient=false, max_properties=800]

+ Thought: Inspecting SpotLight component · 10.8s

⚙unreal_unreal_inspect_object [object_path=/Game/ToggleLight.ToggleLight:PersistentLevel.BP_ToggleLight_C_1.SpotLight, include_transient=false, max_properties=500]

+ Thought: Troubleshooting box activation · 12.7s

+ Thought: Checking help text visibility · 797ms

⚙unreal_unreal_inspect_object [object_path=/Game/ToggleLight.ToggleLight:PersistentLevel.BP_ToggleLight_C_1.HelpText, include_transient=false, max_properties=300]

+ Thought: Providing Fixes for Input Events · 8.7s

+ Thought: Crafting Response for Toggle Light Blueprint · 1.8s

Your Blueprint graph wiring is fine (Overlap Begin → Enable Input → show text, F Pressed → Toggle Visibility on SpotLight). The reason nothing happens is your components are not active at runtime.

From the placed instance (BP_ToggleLight_C_1) I can see:

- Box (trigger) has bAutoActivate = False (so overlap events typically won't fire, meaning Enable Input never runs)

- SpotLight has bAutoActivate = False and starts bVisible = False

- HelpText also has bAutoActivate = False

Fix (in BP_ToggleLight Components panel)

1. Click Box → Details → Activation → enable Auto Activate (and make sure Generate Overlap Events stays on).

2. Click SpotLight → Details → Activation → enable Auto Activate.

3. (Optional) Click HelpText → enable Auto Activate too.

If it still doesn't work after that

Your overlap uses Cast To BP_ThirdPersonCharacter. If your player pawn isn't exactly that class, the cast fails and you never enable input. In that case, remove the cast, or replace it with OtherActor == GetPlayerCharacter(0) before enabling input.

▣ Build · GPT-5.2 · 1m 35s

there is no auto activate option in details

+ Thought: Troubleshooting Unreal Engine Components · 6.9s

+ Thought: Evaluating Overlap and Input Issues · 2.9s

+ Thought: Debugging Overlap Events · 3.4s

+ Thought: Checking Visibility and Overlap Issues · 15.6s

+ Thought: Investigating Input and Overlap Issues · 3.0s

⚙unreal_unreal_get_blueprint_node [object_path=/Game/BP_ToggleLight.BP_ToggleLight, graph=EventGraph, node_id=780891DB-41CF-B571-705E-2EBB12751ED5]

⚙unreal_unreal_get_blueprint_node_defaults [object_path=/Game/BP_ToggleLight.BP_ToggleLight, graph=EventGraph, node_id=780891DB-41CF-B571-705E-2EBB12751ED5]

⚙unreal_unreal_get_blueprint_node [object_path=/Game/BP_ToggleLight.BP_ToggleLight, graph=EventGraph, node_id=3EAA6D92-4284-EAD6-E93F-E78AA04A7620]

⚙unreal_unreal_get_blueprint_node_defaults [object_path=/Game/BP_ToggleLight.BP_ToggleLight, graph=EventGraph, node_id=3EAA6D92-4284-EAD6-E93F-E78AA04A7620]

⚙unreal_unreal_get_blueprint_node [object_path=/Game/BP_ToggleLight.BP_ToggleLight, graph=EventGraph, node_id=F40C2155-4829-18EE-817D-4E83DC06E418]

⚙unreal_unreal_get_blueprint_node_connections [object_path=/Game/BP_ToggleLight.BP_ToggleLight, graph=EventGraph, node_id=F40C2155-4829-18EE-817D-4E83DC06E418]

+ Thought: Debugging blueprint connections · 11.8s

+ Thought: Identifying spotlight issues · 3.1s

+ Thought: Checking spotlight and overlaps · 1.8s

+ Thought: Checking spotlight visibility and collisions · 1.3s

There doesn't need to be an "Auto Activate" option for this to work (and on some components you won't see it). Your problem is almost certainly one of these three things:

1. Your overlap isn't firing, so Enable Input never runs.

- When you walk up to the lamp, do you see the HelpText ("Press F to toggle") appear?

- If you do not see the text: select the Box component in BP_ToggleLight and make sure:

- Collision Enabled = Query Only (or Query and Physics)

- Generate Overlap Events = true

- Collision Preset = OverlapAllDynamic (or overlaps Pawn)

- Increase BoxExtent so you definitely walk into it

- Temporarily unhide it: set Hidden In Game = false so you can see where the box is

2. The cast is failing, so overlap fires but the logic stops.

- Your graph casts OtherActor to BP_ThirdPersonCharacter. If your player pawn is a different class, the cast fails and you never enable input.

- Quick fix: replace the cast with OtherActor == GetPlayerCharacter(0) (Branch), then Enable Input.

3. F input is being consumed elsewhere, so your actor never receives the key even after Enable Input.

- If your character/controller already uses F (common), it may consume the input first.

- Fix: click the F input node in the lamp Blueprint and in the Details panel uncheck Consume Input (and/or raise the actor Input Priority in Class Defaults).

If you tell me whether the “Press F to toggle” text appears when you approach, I’ll narrow it to the exact fix.

▣ Build · GPT-5.2 · 59.9s

Success. the collision box was too small, i managed to see the "Press F to toggle" and also managed to toggle the lamp on and off.

Good. That confirms the input logic was fine; the overlap just wasn’t happening because the Box extent didn’t reach the player.

If you want it to feel solid:

- Increase BoxExtent a bit more than “just barely works” (gives players a buffer).

- Keep HelpText hidden by default (Hidden In Game true), then show it on begin overlap and hide on end overlap (which you already have).

▣ Build · GPT-5.2 · 4.0s